

A General Capacity Frontier of Complex Networks

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Abstract

From cells to cities, ecosystems to economies, complex networks exhibit peak intensities that define their empirical limit. Estimating these limits traditionally requires domain-specific knowledge. We demonstrate that a wide range of complex networks share a capacity frontier that constrains these maxima. We formalize this regularity by proposing Structural Capacity Theory for Network Substrate Structured Systems. These systems couple a fixed network substrate with an observable dynamical process. The frontier is learned from a vector representation of the network substrate and modulated log-additively by a vector representation of the dynamical process. Frontiers trained on one set of domains reliably bound empirical maxima of entirely different ones without observing their peak intensities or simulating their dynamics. Independent learning paradigms converge on consistent upper bounds. Frontier performance is stable under perturbation, yet degrades under deliberate falsification, with no performance gain under ablation. These findings support the capacity frontier as a generalizable constraint that describes how network structure bounds dynamical processes across otherwise unrelated systems.

Maxima are critical for understanding complex networks. Network maxima are important because they are boundary characteristics where critical thresholds are met that mark changes in behavior [1–5]. For example, a regulatory network that exceeds its threshold of localized molecular activity changes regimes [2, 6]. A transportation network that exceeds its carrying capacity changes behavior primarily by congestion, not by design [7, 8]. Although it is known that maxima can affect network behavior, understanding *how* traditionally requires domain-specific knowledge [2, 7]. Whether or not network maxima express a common regularity across domains is still an open question [9–11]. Addressing it requires a shared basis of comparison.

In the previous examples, network structure formed the basis on which a dynamical process occurred [2, 4, 7, 8, 11, 12]. Size, connectivity, spectral organization, and other topological characteristics constrain network throughput, propagation, synchronization, etc. [13–22]. The same network substrate can have different limits depending on the observational definition, activity attribution, temporal aggregation, and measurement regime [23–25]. Conversely, structurally distinct substrates and dynamical processes can exhibit comparable limits under shared measurement conventions and operational definitions [11, 26, 27]. In both cases, empirical maxima reflect the network substrate and the dynamical process by which observations are attributed, generated, and measured.

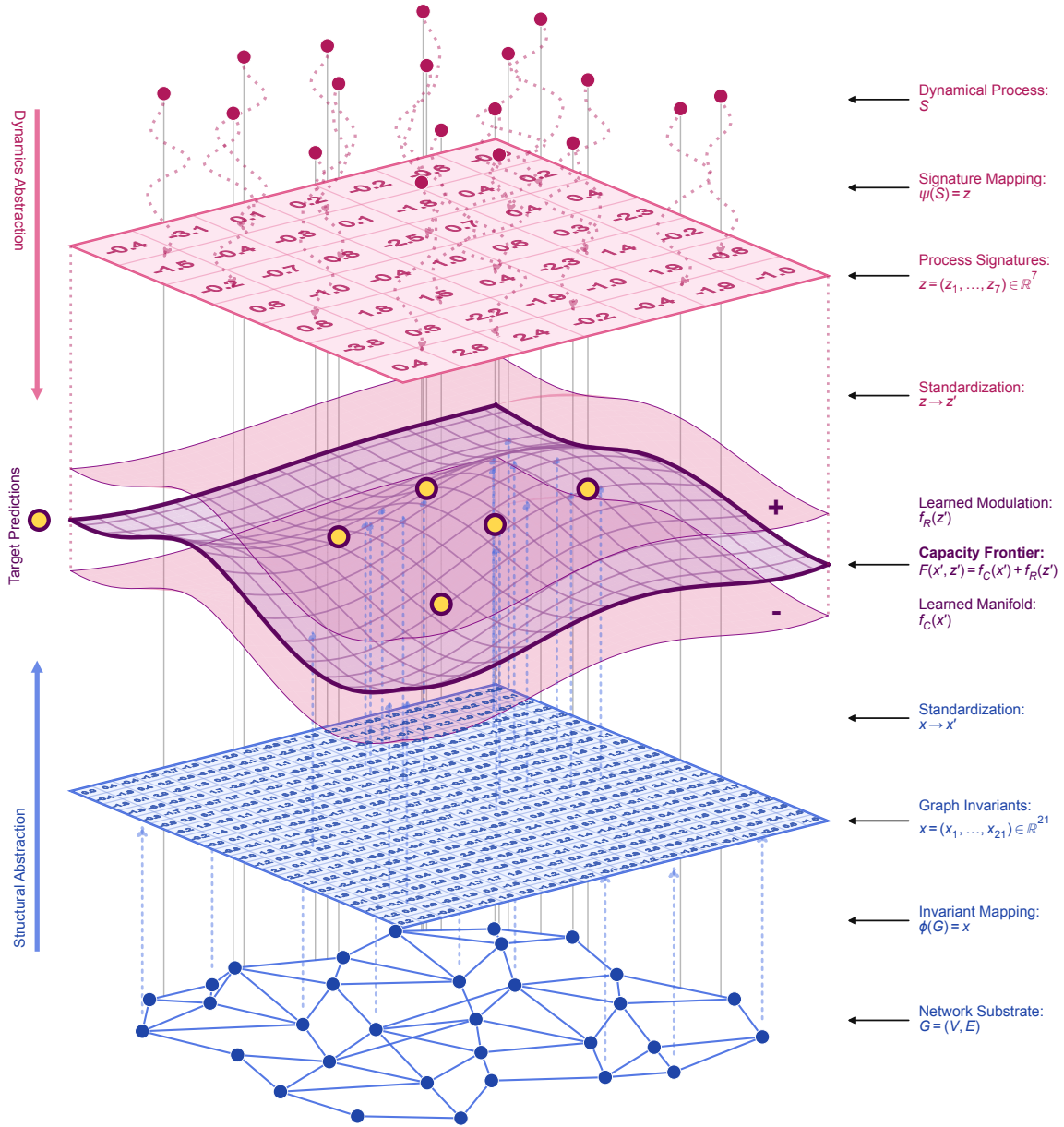


Figure 1: **Conceptual representation of Structural Capacity Theory (SCT)**. Schematic overview of the Network Substrate Structured Systems (N3S) framework as it relates to the capacity frontier. The network substrate G (blue graph on bottom) is encoded as a standardized graph invariant vector x' on the structural plane. The dynamical process S (pink markers on top) produces observations attributed to substrate elements and is encoded as a standardized process signature vector z' on the dynamics plane. The curved manifold represents the capacity frontier $F(x', z') = f_C(x') + f_R(z')$, where the fitted structural component $f_C(x')$ is modulated log-additively by the fitted process component $f_R(z')$ to estimate an upper envelope of empirical maxima y^* (gold points in center).

We propose Network Substrate Structured Systems (N3S) as a class of systems for which this coupling can be evaluated. An N3S is defined by a fixed network substrate, an observable dynamical process that corresponds to that substrate, and a measurable maximum that corresponds to that dynamical process (Fig. 1). Membership is determined from the observations a given system naturally provides so that the scope of the theory is fixed before it is tested. Structural Capacity Theory (SCT) proposes that empirical maxima from *any* N3S are comparable when represented in a shared coordinate system, and within that representation, maxima are constrained by a common capacity frontier.

We estimate the capacity frontier as a covariate-dependent upper envelope. Typical network activity can be captured by a conditional mean, but the quantity of interest is the ceiling, so the estimators are trained as such [28–34]. The system’s network structure and dynamical process are encoded as standardized vector representations x' and z' , respectively. The capacity frontier $F(x', z') = f_C(x') + f_R(z')$ log-additively bounds the log-transformed maximum $y^* = \ln(1 + r_{\max})$. The frontier is strictly empirical. It permits a small fraction of observations to cross it and carries no claim of analytical containment.

We evaluate whether this formulation supports a transferable, consistent, stable, falsifiable, and decomposable capacity frontier. We found a frontier that bounded empirical maxima from out-of-sample systems using Leave-One-Group-Out Cross-Validation (LOGO-CV) and repeated k -Fold Cross-Validation (k -Fold-CV) with equivalent performance. Independent estimators from 3 learning paradigms predicted highly consistent frontier values. Perturbations to network substrates, graph invariants, dynamical processes, and process signatures preserved frontier performance. By contrast, the 3 falsification procedures that deliberately destroyed the mapping from (x', z') to y^* degraded frontier performance that retraining could not recover. Finally, ablations that tested single components and richer couplings did not meaningfully improve the performance of the frontier.

Shared Coordinate System

Encoding all systems as paired network substrate and dynamical process vectors placed unrelated phenomena within a shared coordinate system (Fig. 1, Supplementary Table 1). The corpus of systems contained 5 broad domains: Earth & Physical Sciences, Life Sciences & Medicine, Technology & Information, Trade & Institutions, and Transportation & Infrastructure. Each of the 5 broad domains contained 5 individual disciplines. Their measurements did not share units before encoding. The network substrate vector x admitted 21 graph invariants and the dynamical process vector z admitted 7 process signatures. Both were selected under prespecified criteria and standardized to x' and z' . This encoding represented every N3S with the same number of dimensions despite large differences in substance and scale. It made empirical maxima from a wide range of domains and magnitudes directly comparable.

Within this shared representation, the capacity frontier bounded empirical maxima across the corpus (Supplementary Table 2). Under repeated 5-Fold-CV and 10-Fold-CV, the frontier was fit on each training set and evaluated on the corresponding test set. Each test was repeated 30 times to reduce dependence on any single seeded partition. Frontier performance was reported as medians across estimators after averaging them over test folds and repetitions. In 5-Fold-CV and 10-Fold-CV, frontier efficiency was 0.77 under both resampling regimes. Frontier efficiency was measured using the Efficiency Index (EI) — the geometric mean of the non-violation rate and two bounded metrics [0, 1] that decreased with violation magnitude and slack relative to the evaluated target scale. There were infrequent frontier violations, low violation magnitude, and consistent slack. The violation rate was 0.03 and 0.02 under 5-Fold-CV and 10-Fold-CV, respectively. Mean violation magnitude was 0.12 and 0.03, and mean slack was 6.54 and 5.68 log units under 5-Fold-CV and 10-Fold-CV, respectively. The formulation was found to accommodate the entire corpus of 25 individual systems.

Transfer Across Domains

A capacity frontier trained without a given domain constrained the empirical maxima of that domain’s disciplines (Supplementary Table 2). Under domain-level LOGO-CV, the frontier was trained on 4/5 broad domains and evaluated on the 5 individual disciplines in the withheld domain. Frontier efficiency was 0.76, ranging from 0.67 to 0.83 (Fig. 2a). Estimation-observation consensus was 0.83, ranging from 0.77 to 0.92 across the same evaluations (Fig. 2b). Estimation-observation consensus was measured using the Consensus Index (CI) — the agreement between estimated and observed maxima measured by the geometric mean of global rank agreement, high-capacity overlap, and nonlinear dependence. The violation rate was 0.04. Mean violation magnitude was 0.21. Mean slack was 6.93 log units. Each withheld discipline supplied its own graph invariants, process signatures, and observed maxima so the protocol tested domain transfer of the frontier specification to a domain absent from training, not prediction in the absence of observation. This established generalization across tested domains.

A capacity frontier trained without a given discipline also bounded that discipline’s empirical maxima. Withholding a single discipline tested similar generalization as domain-level LOGO-CV, but at a higher resolution. Under discipline-level LOGO-CV the frontier was trained on 24/25 disciplines and evaluated on the one that was withheld, effectively Leave-One-Out Cross-Validation (LOO-CV). In this resampling regime frontier efficiency was 0.73, ranging from 0.52 to 0.87 across disciplines. Estimation-observation consensus was 0.83 based on out-of-fold predictions pooled within each of the 5 broad domains, ranging from 0.79 to 0.90 across domains. The violation rate was 0.01. Mean violation magnitude was 0.01. Mean slack was 5.69 log units. The violation rate was approximately a quarter of domain-level LOGO-CV, and mean violation magnitude was approximately an order of magnitude smaller. The experiment could not determine whether these differences reflected denser coverage of the coordinate system by the 24 in-sample disciplines or differences in test-fold size and metric aggregation. Nevertheless, the frontier generalized to out-of-sample disciplines similarly to out-of-sample domains, indicating indifference to resampling resolution.

The capacity frontier achieved equivalent frontier performance across resampling regimes. Using the repeated k -Fold-CV results from **Shared Coordinate System**, paired differences in frontier efficiency between domain-level LOGO-CV and repeated k -Fold-CV were tested across disciplines within the 5 broad domains (Fig. 2c). Paired differences were computed by subtracting domain-level LOGO-CV outcomes from those contained in the comparative resampling regime, where negative values indicated lower performance. Median paired differences in frontier efficiency were 0.01 at most for all resampling regimes. Median paired differences in estimation-observation consensus were 0.01 at most under the same evaluation. Paired Wilcoxon Two One-Sided Tests (TOST) across 45 estimator-by-domain comparisons supported equivalence within their empirically derived margins ($\delta_{EI} = 0.01$ and $\delta_{CI} = 0.03$, all Holm-adjusted $p \leq 0.042$, Supplementary Table 3). Withholding broad domains and individual disciplines preserved frontier performance when compared to repeated random resampling. Thus, domain generalization did not depend on any single tested resampling regime.

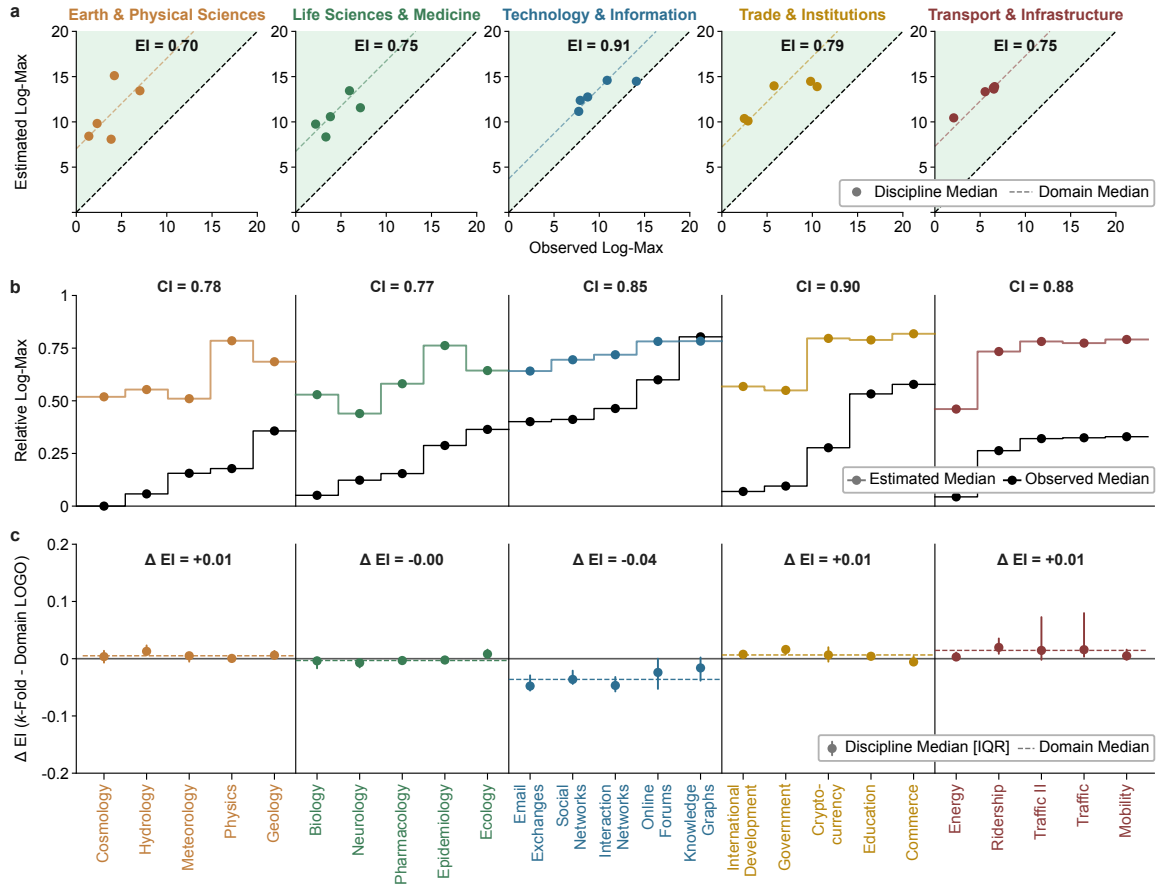


Figure 2: **Generalization of the capacity frontier.** Domain-level LOGO-CV across 25 disciplines in 5 broad domains. **a.** Estimated versus observed log-maxima for each withheld domain (top columns). Discipline medians (domain-colored markers) fall along domain offsets (domain-colored dashed lines) relative to equality (gray dashed diagonal). Green shading marks conservative estimates at or above observed maxima. **b.** Estimation-observation consensus profiles within withheld domains (middle row). Step traces and markers show observed log-maxima (black trace on bottom) and median model estimates (domain-colored trace on top), with disciplines ordered by observed maximum and profiles scaled to the unit interval. **c.** Paired differences in frontier efficiency between random cross-validation (5-Fold-CV and 10-Fold-CV) and domain-level LOGO-CV (random – LOGO, bottom row). Markers and vertical intervals show discipline medians and interquartile ranges across models (domain-colored intervals), dashed lines show domain medians (domain-colored dashed lines), and the horizontal band spans the interquartile range of baseline variation from model means (light blue band). Annotations report domain EI (**a**), domain CI (**b**), and domain-median ΔEI (**c**).

Consensus Across Paradigms

The capacity frontier was consistent across learning paradigms (Fig. 3). 9 independent estimators across 3 distinct learning paradigms agreed on their predicted frontier values. The learning paradigms were linear parametric, non-linear ensemble, and neural networks. All 9 models were trained on the same vectors x' and z' while the estimators differed in functional form, regularization, and optimization. Each test was repeated 30 times to reduce sensitivity to parameter initialization and

optimization stochasticity. Predicted frontier values were averaged across repetitions. Pairwise model consensus across all 36 pairs of seed-averaged predictions was 0.84. A geometric mean this close to 1 simultaneously required high rank agreement, top-weighted overlap, and statistical dependence, such that no single component could carry the outcome. Spearman rank correlation, rank-biased overlap, and distance correlation were 0.82, 0.85, and 0.80, respectively. The agreement was evaluated on in-sample analyses using full-corpus fits. Consensus across the tested learning paradigms indicated that the frontier reflected a regularity, which was unlikely an artifact of inductive bias.

Stable Under Perturbation

The capacity frontier was robust to perturbation (Fig. 4a, Fig. 4b, Supplementary Table 4). The 12 perturbations targeted 4 categories: network substrates, graph invariants, dynamical processes, and process signatures. Under domain-level LOGO-CV, the frontier was trained on the original data, the model frozen, then evaluated on the perturbed data in the withheld domain. Each test was repeated 30 times and averaged across repetitions to reduce sensitivity to perturbation realization and estimator initialization. Paired differences were computed by subtracting the original outcomes from the perturbed ones across 45 estimator-by-domain comparisons, where negative values indicated lower performance. At the highest tested intensity, median paired differences in frontier efficiency ranged from -0.02 to +0.01. Median paired differences in estimation-observation consensus were 0.00 for all network and invariant perturbations. Median paired differences in consensus were -0.04 for process bootstrapping, -0.02 for process scaling, and -0.06 for process smoothing. Signature jitter, noise, and masking produced median consensus differences of -0.01, -0.03, and -0.06, respectively. Paired Wilcoxon Two One-Sided Tests (TOST) across 45 estimator-by-domain comparisons supported equivalence for all 12 perturbations within their empirically derived margins ($\delta_{EI} = 0.09$ and $\delta_{CI} = 0.15$, all Holm-adjusted $p < 0.001$). Response curves across normalized perturbation intensity were descriptive, where only the highest perturbation intensity was formally tested. Nevertheless, the frontier could withstand bounded structural alterations, process distortions, and feature noise while remaining reliable.

Degradation Under Falsification

The capacity frontier predictably degraded under falsification (Fig. 4c, Supplementary Table 5). There were 3 falsification procedures that disrupted the correspondence between feature representations and target maxima in distinct ways: target permutation preserved the empirical target distribution, random generation sampled within observed feature ranges, and vector generation maintained approximate featurewise scale. Under domain-level LOGO-CV, the frontier was evaluated by freezing the models after training on the original data, and separately after retraining on the falsified data. Each test was repeated 30 times and averaged across repetitions to reduce sensitivity to estimator initialization and training stochasticity. Paired differences were computed by subtracting the original outcomes from the falsified ones across 45 estimator-by-domain comparisons, where negative values indicated lower performance. Under the frozen model, median paired differences in frontier efficiency were -0.02 for target permutation, -0.05 for random generation, and -0.03 for vector generation. After retraining, median paired differences in frontier efficiency were -0.08 for target permutation, -0.03 for random generation, and -0.03 for vector generation. Median paired differences in estimation-observation consensus were -0.24, -0.24, and -0.21 when frozen, and -0.33, -0.21, and -0.28 after retraining, respectively. Paired One-Sided Wilcoxon Signed-Rank Tests supported lower efficiency and lower consensus across all 12 comparisons (all Holm-adjusted $p \leq 0.005$ for efficiency and $p < 0.001$ for consensus). The falsified corpora served as prespecified negative controls. They indicated that the frontier depended on a true representation–target correspondence.

Parsimony Under Ablation

No alternative capacity frontier specification meaningfully outperformed the log-additive decomposition (Fig. 4d, Supplementary Table 6). The 4 alternative specifications contained simple ablations: invariants only and signatures only, and complex ablations: joint features and interaction terms. Invariants only trained the frontier using x' alone, while signatures only used z' . Joint features combined both vectors into a single-stage fit, while interaction terms augmented the log-additive decomposition with pairwise products between the 2 vectors. Under domain-level LOGO-CV, ablations were fit on each training set and evaluated on the corresponding test set. Each test was repeated 30 times and averaged across repetitions to reduce sensitivity to estimator initialization and training stochasticity. Paired differences were computed by subtracting outcomes from the original formulation from the ablated ones across 45 estimator-by-domain comparisons, where negative values indicated lower performance. Median paired differences in frontier efficiency were -0.01 for invariants only, joint features, and interaction terms, and 0.00 for signatures only. Median paired differences in estimation-observation consensus were -0.30 for invariants only, -0.09 for signatures only, -0.26 for joint features, and -0.08 for interaction terms, respectively. Paired One-Sided Wilcoxon Signed-Rank tests supported non-inferiority of the log-additive decomposition to all 4 alternatives in both metrics within their empirically derived margins ($\delta_{EI} = 0.07$ and $\delta_{CI} = 0.14$, all Holm-adjusted $p < 0.001$). These tests limited alternative advantages. They supported the log-additive decomposition as the parsimonious formulation of the frontier.

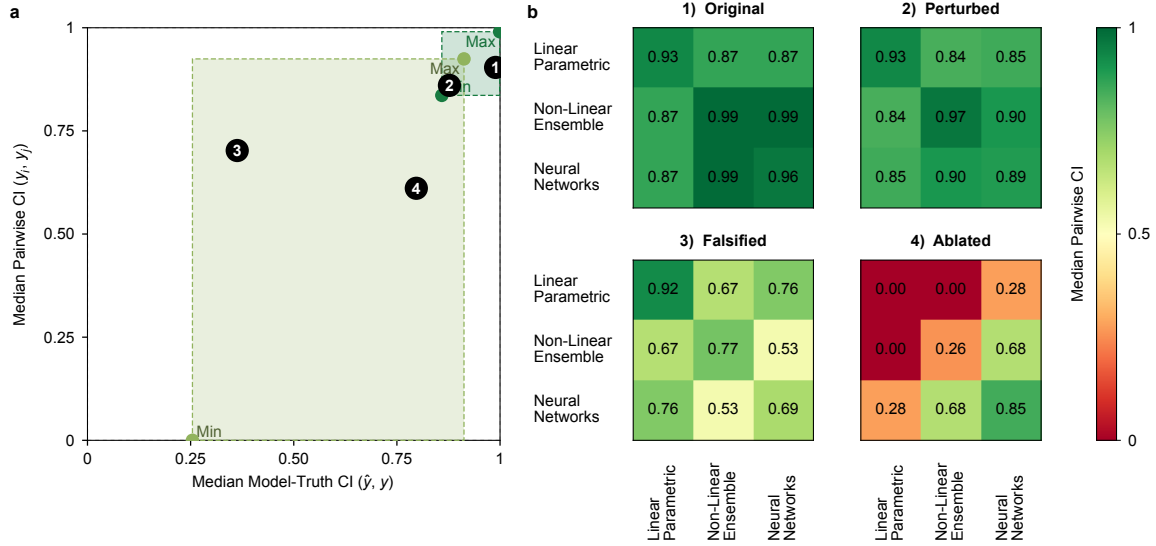


Figure 3: **Learning paradigm and estimation-observation consensus.** Agreement across 9 estimators in 3 learning paradigms evaluated on full-corpus fits across 30 seeds. **a.** Median estimation-observation consensus ($\hat{y}^*$ versus y^*) by median pairwise model consensus ($\hat{y}_i^*$ versus $\hat{y}_j^*$). Points mark the tests: (1) original, (2) perturbed, (3) falsified, and (4) ablated. Shaded rectangles span coordinate-wise ranges of paradigm medians for the original and perturbed (green) and falsified and ablated (yellow) groups. **b.** Median pairwise model consensus across learning paradigms for each test in **a** with paradigm agreement along the diagonal axes.

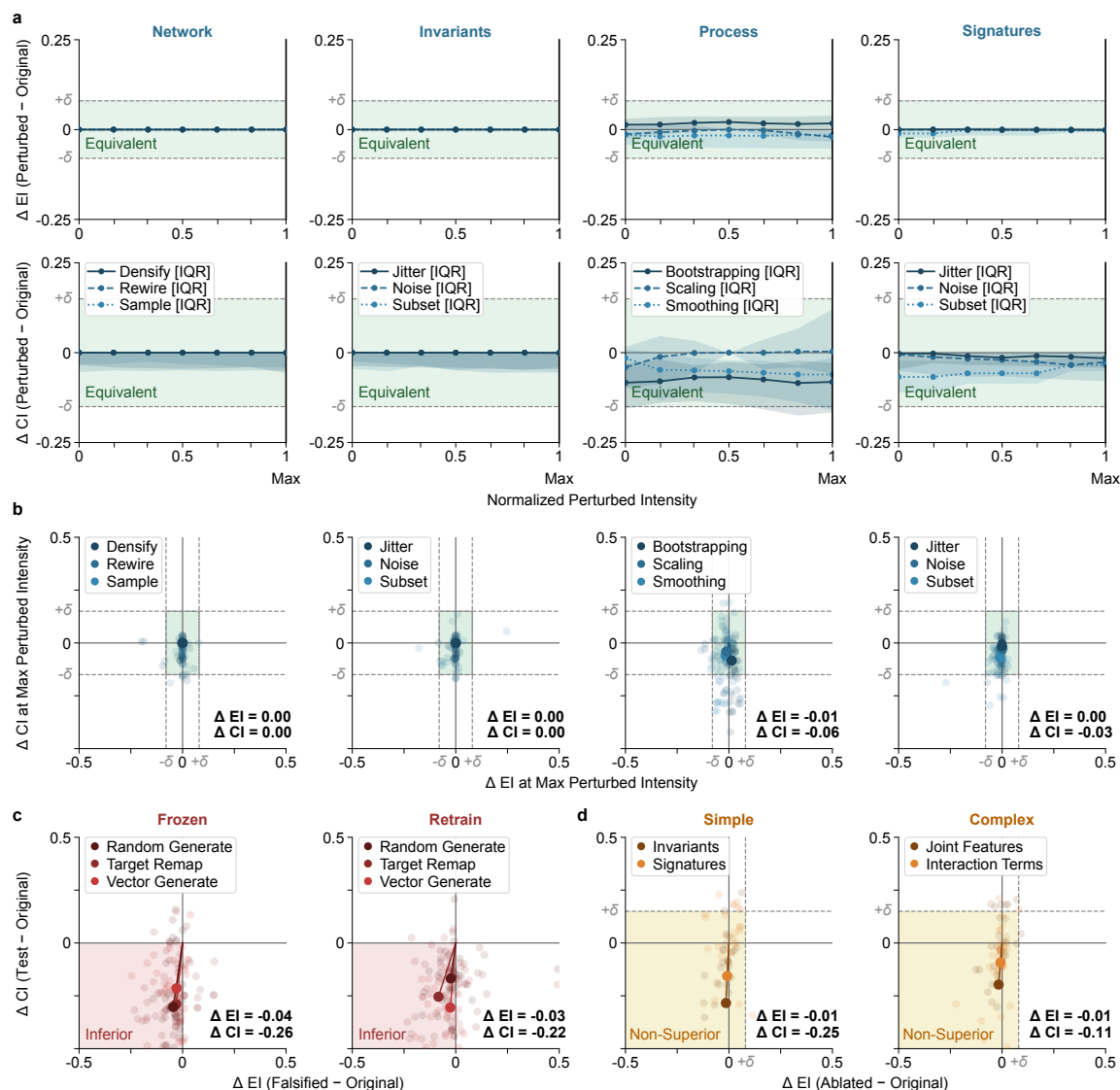


Figure 4: **Stress tests of the capacity frontier.** Paired shifts across 45 estimator-by-domain comparisons under domain-level LOGO-CV, computed as tested condition minus original condition, or alternative specification minus log-additive decomposition. **a.** Response curves across normalized intensity for 12 perturbations in 4 categories, with ΔEI above and ΔCI below. Lines and shaded bands show medians and interquartile ranges. **b.** Paired shifts at maximum perturbation intensity. **c.** Falsification shifts under frozen and retrained frontiers for target permutation, random generation, and vector generation. The red quadrant marks joint degradation ($\Delta EI < 0$, $\Delta CI < 0$). **d.** Simple (invariants only, signatures only) and complex (joint features, interaction terms) specifications compared with the log-additive decomposition. In **b–d**, faint points show individual comparisons and solid points show coordinate-wise medians connected to the origin. Green regions (**a**, **b**) mark perturbation equivalence margins ($\delta_{EI} = 0.09$, $\delta_{CI} = 0.15$), and the yellow region (**d**) marks the ablation non-inferiority margin ($\delta_{EI} = 0.07$, $\delta_{CI} = 0.14$).

Discussion

The capacity frontier makes empirical maxima comparable across disciplinary boundaries. A frontier with either a broad domain or individual discipline withheld from training bounded empirical maxima with performance comparable to random resampling. Information from one system was therefore useful for estimating the maxima in another despite differences in subject, size, scale, units, and dynamics. These comparisons provide a basis for asking which constraints are shared across systems and which depend on domain specificity. Capacity estimation can therefore use empirical evidence beyond a system’s own substantive boundary.

The capacity frontier does not require a model of the dynamical processes. Graph invariants and process signatures support capacity estimation without training domain specific models or simulating their dynamics. Domain specific models are still important for causal explanation and intervention, but SCT provides a standard benchmark as a complement. Interpreting this benchmark requires additional inquiry, as the reported slack above the maxima was nearly three orders of magnitude. This gap may reflect a loose statistical envelope, unequal observation windows, or differences in measurement conventions. Although the meaning of the slack remains open, the capacity frontier still placed most evaluated maxima below its estimates without a traditional model of the underlying dynamics.

Selective stability demonstrates why the capacity frontier is informative. Perturbations at the highest tested intensity generally preserved the frontier’s original performance. Falsifications that deliberately destroyed the representation–target correspondence degraded frontier performance, which was unrecoverable even after retraining. Consensus among learning paradigms was relatively high and comparatively similar between the original and perturbed conditions. Only after falsification and ablation did the model consensus between learning paradigms degrade. This supports the frontier as a non-spurious object, distinguishable from a flexible upper envelope.

The capacity frontier is jointly determined and decomposable. Removing either component or combining them without log-additivity did not improve frontier performance or consensus among estimators. The results support the log-additive decomposition, but leave its exhaustiveness unresolved. All findings were conditioned on the 5 broad domains, 21 graph invariants, 7 process signatures, 3 learning paradigms, the log-additive decomposition, and the prespecified daily aggregation. Broader corpora, different vector representations, more diverse learning paradigms, and alternative temporal scales all provide direct tests of SCT.

Methods

N3S Definition

Network Substrate Structured Systems (N3S) comprise a fixed graph substrate G , an observable dynamical process, and a convention for measuring an empirical rate maximum. The substrate contains finite elements with relational or attribution structure. The process is a time-indexed sequence of observed events or events reconstructed from published canonical timings, each attributable to substrate elements. The convention converts the sequence into an empirical rate series.

Membership required empirical records for all three components, with operational recastings fixed before evaluation (Supplementary Table 1). Systems without an identifiable substrate or with only aggregate observations lacking substrate attribution were excluded.

Data and Corpus Composition

The corpus contained 25 systems from 25 distinct disciplines, five within each domain: Earth & Physical Sciences, Life Sciences & Medicine, Technology & Information, Trade & Institutions, and Transportation & Infrastructure. Public data sources, access parameters, operational recastings, and system definitions appear in Supplementary Table 1.

Network substrates ranged from 5 to approximately 10,400,000 nodes and from 10 to approximately 7,200,000,000 edges. Empirical maximum rates spanned from 3 to roughly 1,300,000 observations per day. Observation windows reflected native recording periods or canonical intervals without standardized durations. Measurements shared no units prior to encoding, and daily aggregation served as the common temporal convention. Each system contributed exactly one analysis record containing its graph invariants, process signatures, and empirical maximum.

Network Substrate Construction

Every network substrate was represented as a static, unweighted, undirected graph:

$$G = (V, E), \quad E \subseteq \binom{V}{2} \quad (1)$$

Vertices represented units, and edges native ties or prespecified associations. Directed and weighted records were converted to undirected, unweighted edges. Multi-edges and self-loops were removed, isolated vertices retained, and substrates held fixed across their observation windows.

An attribution map $a : \mathcal{O} \rightarrow V \cup E$ linked each observation $o \in \mathcal{O}$ to a vertex or edge through a measured or prespecified correspondence. Interaction logs formed unions of observed pairs or complete bipartite graphs over documented entity classes, including entity-time constructions. Four complete bipartite substrates (college, email, mooc, and overflow) were defined by partition sizes m and n . Their invariants used exact closed-form expressions for $K_{m,n}$ to avoid instantiating billions of edges (Supplementary Table 7). Supplementary Table 1 defines each system’s nodes and edges.

Dynamical Process Construction

Dynamical processes were derived directly from empirical observation records. For continuous records, measurements were discretized using the prespecified operational thresholds documented in Supplementary Table 1. Where only population-mean transition times were published, records were structured as canonical event sequences. Event counts were aggregated into uniform daily intervals using UTC timestamps when present, or consecutive 24-hour bins for relative timestamps and regular snapshots. This aggregation produced a discrete daily rate series:

$$S = \{r_t\}_{t \in T}, \quad r_t = \frac{n_t}{\Delta t}, \quad \Delta t = 1 \text{ day (24 hours)} \quad (2)$$

where n_t is the event count in interval t and T is the chronologically ordered set of evaluated intervals.

The composition of the interval set T followed source conventions. For 19 systems, T contained only days with at least one recorded event, omitting event-free days. Five systems (metrila, pemsbay, montevideo, windmill, and wiki) supplied regular snapshots that aggregated into complete daily series. Chickenpox supplied weekly county totals, which were divided by seven to establish daily averages and entered as single series values at their report dates. Supplementary Table 1 lists the convention for each system.

The full observed rate series supplied both the peak target and seven process signatures. In held-out folds, models used graph invariants from G and signature summaries from that series, without the target scalar or raw trajectory as inputs. The Fano factor retained count scale: its ratio to the squared coefficient of variation equaled the series mean $\bar{r}$. The other six signatures were scale-free. No fitted dynamical equations or generative simulations entered the construction.

Empirical Maximum Target

The estimation target for each system was the log-transformed maximum daily rate realized across its empirical observation window. For each rate series S on interval set T , the peak daily rate was $r_{\max} = \max_{t \in T} r_t$. A sample maximum can grow with observation-window length [35–37]. Observation windows varied across sources without an artificial horizon, so all capacity frontiers remained conditional on the empirical windows in Supplementary Table 1. The estimation target was the shifted log-transformed maximum:

$$y^* = \ln(1 + r_{\max}) \quad (3)$$

Each system contributed exactly one scalar target value.

The logarithmic transformation mapped maxima spanning five orders of magnitude onto a shared scale, with the unit shift accommodating zero counts. Slack and violation magnitudes were expressed in natural-log units. Violation rate, normalized mean violation, normalized mean slack, the Efficiency Index, and the Consensus Index were dimensionless. The capacity frontier $F(x', z')$ operated as an empirical upper envelope over these finite-window targets within the shared feature space. Envelope performance was measured directly by violation metrics without asserting analytical containment across all dynamical regimes on G .

Encoding and Standardization

Each substrate contained 21 graph invariants in five families: cardinality and cut structure, cohesion, extremal statistics, degree distributions, and spectral moments:

$$x = \phi(G) = (x_1, x_2, \dots, x_{21}) \in \mathbb{R}^{21} \quad (4)$$

Each chronologically ordered series contained seven process signatures spanning temporal dependence, dispersion, local volatility, long-range dependence, trend, and distributional asymmetry:

$$z = \psi(S) = (z_1, z_2, \dots, z_7) \in \mathbb{R}^7 \quad (5)$$

All features met prespecified criteria: classical, global to the network or series, free of arbitrary thresholds or external parameters, deterministic at corpus scale, and finite for all 25 systems. Supplementary Tables 7 and 8 provide definitions, citations, and asymptotic complexities.

Coordinates were standardized (centered and scaled) using training-fold population standard deviations, with unit scale for zero-variance coordinates:

$$x' = \frac{x - \mu_x}{\sigma_x}, \quad z' = \frac{z - \mu_z}{\sigma_z} \quad (6)$$

Means and scales were fitted exclusively on training rows within each fold.

Capacity Frontier Learning

The capacity frontier was learned through a two-stage log-additive architecture with prespecified estimator settings. Each stage received at most 21 or 7 predictors without estimator tuning. Thirty random seeds assessed numerical stability, and nine estimators across three distinct learning paradigms evaluated sensitivity to inductive bias.

The fitted frontier took a stagewise log-additive form with an identifiability constraint on the process stage [38]:

$$\hat{y}^* = F(x', z') = f_C(x'; \tau_C) + f_R(z'; \tau_R) \quad (7)$$

Stage 1 fit the structural component f_C to target y^* via asymmetric regression at $\tau_C = 0.99$, placing the fit near the empirical upper envelope given graph invariants [34]. Stage 2 fit the process component f_R to the signed slack $y^* - \hat{f}_C(x')$ at $\tau_R = 0.50$, modeling how process signatures modulated the structural ceiling. Subtracting the training mean of f_R assigned the overall level to f_C and centered modulation to f_R . Summing both stages yielded the frontier estimate. The structural level τ_C set stage asymmetry. Joint envelope coverage was evaluated empirically through violation rate, violation magnitude, slack, and efficiency.

Nine estimators instantiated this form with identical asymmetry levels and fixed hyperparameters. The linear parametric paradigm included three models. Linear quantile regression minimized mean pinball loss with $\alpha \|\beta\|_1$ penalty ($\alpha = 0.1$, HiGHS solver) [31]. Linear asymmetric weighted least squares minimized $(2n)^{-1} \sum_{i=1}^n w_i e_i^2 + \alpha \|\beta\|_1$ ($\alpha = 0.1$) with expectile weights $w_i = \tau$ for $e_i \geq 0$ and $w_i = 1 - \tau$ otherwise [39]. Coefficients started at zero, and cyclic coordinate descent used tolerance 10^{-4} and at most 1,000 passes [40]. Stage 1 used an unpenalized intercept initialized at the training-target mean, and stage 2 used no intercept. The concave-envelope regressor fit a linear quantile score ($\alpha = 0.1$) and linearly interpolated the concave upper envelope of target maxima along those scores, holding it constant beyond endpoints.

The non-linear ensemble paradigm included a quantile random forest (100 fully grown trees) [41], gradient-boosted quantile regression (100 trees, maximum depth 3, learning rate 0.1) [42], and XGBoost under its quantile objective (100 trees, maximum depth 6, learning rate 0.3, ℓ_2 penalty $\lambda = 1$) [43]. The neural network paradigm included an MLP with pinball loss, an MLP with expectile loss, and an input-convex network with pinball loss whose non-input weights were constrained non-negative through absolute values [44]. All networks employed hidden layers of 8 and 4 units, ReLU activations [45], dropout of 0.1 [46], output biases initialized to the training quantile, and full-batch AdamW optimization (learning rate 0.1, weight decay 0.01, gradient clipping at 5.0, 1,000 epochs) [47], with predictions clipped to the training-target range.

Performance Metrics

Frontier tightness was summarized by the Efficiency Index (EI) [48, 49]. For observed targets y^* and frontier values $\hat{y}$, the violation rate VR was the fraction of systems with $y^* > \hat{y}$, the mean violation MV was the average exceedance among violators (0 when none occurred), and the mean slack MS was the average positive difference $\max(\hat{y} - y^*, 0)$ across all evaluated systems. EI was the geometric mean of three bounded scores, normalizing MV and MS by the sample mean absolute target $|y^*| = n^{-1} \sum_{i=1}^n |y_i^*|$:

$$\text{EI} = \left[(1 - \text{VR}) \left(1 + \frac{\text{MV}}{|y^*|} \right)^{-1} \left(1 + \frac{\text{MS}}{|y^*|} \right)^{-1} \right]^{1/3} \in [0, 1] \quad (8)$$

EI reached 1 only when the frontier exhibited zero violations and zero slack, with $1 - \text{VR}$ floored at 10^{-12} for numerical stability.

Vector agreement was quantified by the Consensus Index (CI). CI was the geometric mean of rescaled Spearman rank correlation $\rho_s^+ = (1 + \rho_s)/2$ [50], top-weighted rank-biased overlap (RBO) [51], and distance correlation (dCor) [52]:

$$\text{CI} = \left[\rho_s^+ \text{RBO dCor} \right]^{1/3} \in [0, 1] \quad (9)$$

RBO evaluated descending rankings with persistence $p_{\text{RBO}} = 0.9$ and index-based tie breaking, extrapolating the unobserved tail from depth agreement. Distance correlation used the standard V-statistic sample estimator. Components were clipped to $[0, 1]$ against precision error, with Spearman correlation set to zero when either vector was constant. A high CI reflected simultaneous global rank concordance, agreement on top-ranking systems, and nonlinear dependence. CI served two roles: estimation-observation consensus between predicted frontiers and empirical maxima, and pairwise model consensus between predictions from different estimators.

Resampling and Leakage Control

Generalization was evaluated under strict leakage controls across four resampling regimes. Standardizers for x and z , both estimator stages, and the process centering mean in Eq. (7) were fitted exclusively on training rows of the active split. Estimators were freshly cloned for every fold. All feature values were finite, and the logarithmic target transform had no fitted parameters.

Random evaluation used repeated 5-Fold-CV and 10-Fold-CV over 30 repetitions, with partitions drawn by `RepeatedKFold` under random state 42 [53]. Structured generalization used leave-one-group-out cross-validation (LOGO-CV) [54], withholding either one broad domain (5 folds) or one discipline (25 folds) [55–58]. Because LOGO partitions were deterministic, the 30 repetitions varied model-fitting seeds $42 + b$ for $b = 0, \dots, 29$. Metrics were computed within each held-out test fold and averaged over folds and repeats. Out-of-fold predictions averaged over the 30 seeds yielded domain-level Consensus Index scores. Under system-level LOGO-CV, predictions were pooled within domains before computing CI. Resampling equivalence rescored averaged out-of-fold predictions within each of the five broad domains.

Stress tests followed two operational tracks. The frozen track applied models and standardizers fitted on original data directly to modified test inputs. The retrained track refit the entire pipeline on modified data under the identical cross-validation protocol. Two corpus-wide operations preceded dataset partitioning: generating the falsified negative control corpora and computing corpus-wide standard deviations for additive feature noise. In all cases, standardizer and estimator parameters remained strictly training-fold specific.

Evaluation and Reporting Conventions

Shared-coordinate frontier quality used repeated 5-Fold-CV and 10-Fold-CV. Domain transfer used domain- and discipline-level LOGO-CV. Resampling equivalence benchmarked domain-level LOGO-CV, perturbation and falsification used the original condition, and ablation used the log-additive decomposition.

Cross-paradigm consensus was evaluated descriptively on full-corpus fits across all nine estimators and 30 seeds. Pairwise model consensus compared the 36 seed-averaged prediction pairs using CI. Figure 3 presents these descriptive full-corpus consensus values across four conditions: original full-corpus fits, frozen fits applied to maximum-severity perturbations across all 25 systems, frozen fits scored on falsified corpora, and refitted ablations. Reported values represented median CI scores across all estimator pairs.

Hypothesis testing operated on a prespecified inference panel $\mathcal{P}$ of 45 estimator-by-domain cells (nine estimators by five withheld domains). Each cell’s metric was averaged across 30 seeds before testing. For metric M , estimator m , and domain d , the paired difference was $d_{md} = M_{md}^{\text{comparison}} - M_{md}^{\text{benchmark}}$, with estimand $\theta_{\mathcal{P}} = \text{median}_{(m,d) \in \mathcal{P}} d_{md}$. Because panel cells shared estimators and data splits, test outcomes co-varied, and statistical inference remained conditional on this panel. All tests employed one-sided Wilcoxon signed-rank tests [59], discarding zero differences under SciPy’s default zero method, with Holm-Bonferroni corrections [60] across each hypothesis family at a family-wise significance threshold of 0.05. Effect sizes were quantified using rank-biserial correlations $(W_+ - W_-)/(W_+ + W_-)$ derived from nonzero differences [61].

Equivalence for resampling regimes and perturbation stability was evaluated using Two One-Sided Tests (TOST) [62]:

$$H_{01} : \theta \leq -\delta, \quad H_{02} : \theta \geq \delta \quad (10)$$

Each null hypothesis was tested on differences shifted by $-\delta$ or $+\delta$. Equivalence was established when both nulls were rejected, with the TOST p -value taken as the maximum of the two tests. The margin δ was the interquartile range of benchmark values, floored to two decimal places. Resampling equivalence used nine model-level domain LOGO means, each averaged across five domains. Perturbation stability used 45 original-condition panel cells, and ablation non-inferiority used 45 log-additive panel cells. The resulting margins were $\delta_{\text{EI}} = 0.01$ and $\delta_{\text{CI}} = 0.03$ for resampling equivalence, $\delta_{\text{EI}} = 0.09$ and $\delta_{\text{CI}} = 0.15$ for perturbation stability, and $\delta_{\text{EI}} = 0.07$ and $\delta_{\text{CI}} = 0.14$ for ablation non-inferiority. Where intervention intensities varied, response curves were displayed descriptively, with formal testing restricted to the maximum severity level documented in Supplementary Table 9.

Perturbation, Falsification, Ablation

Model stability and representation dependence were evaluated through perturbation, falsification, and ablation stress tests. Perturbation tests assessed stability across 12 interventions distributed evenly over four categories: network substrates, graph invariants, dynamical processes, and process signatures. Each intervention was applied to every system across seven graded intensity levels (Supplementary Table 9).

Network interventions included degree-preserving rewiring [63] with $\max(1, \lfloor s|E| \rfloor)$ trials while preserving a simple graph (no self-loops or multiedges), random edge densification [64] adding $\lfloor s|E| \rfloor$ edges sampled uniformly without replacement from nonadjacent pairs, and uniform node removal [65, 66] deleting $\lfloor s|V| \rfloor$ vertices. Invariants were recomputed after each modification. Complete bipartite substrates received exact node removal and closed-form approximations for rewiring and densification (Supplementary Note 1). Invariant and signature perturbations acted directly on raw vectors x and z prior to standardization. Additive Gaussian noise used $\mathcal{N}(0, (s\sigma_j)^2)$, where σ_j was the corpus-wide sample standard deviation of raw coordinate j . Multiplicative jitter multiplied coordinates by $1 + \varepsilon_j$ with $\varepsilon_j \sim \mathcal{N}(0, s^2)$ and clipped originally nonnegative feature columns at zero. Coordinate masking retained $\lfloor \rho p \rfloor$ features and set unselected coordinates to zero, where ρ was the retention fraction and p the feature count (21 for x , 7 for z).

Process interventions acted directly on the rate series of Eq. (2) through power-law scaling $r'_t = (r_t + 1)^\gamma - 1$, smoothing with a w -interval trailing mean, and bootstrap resampling at support s [67]. Bootstrap resampling drew $k = \max(1, \text{round}(s|T|))$ values with replacement and populated the remaining intervals by resampling from those k values, destroying temporal order. Process signatures were recomputed from each modified series, while estimation targets remained fixed at the original empirical maxima.

Falsification tests assessed degradation under three negative control procedures that deliberately destroyed the mapping from feature representations to empirical maxima. Target permutation

assigned each system the complete event series and target y^* of a different system via a cross-domain derangement, preserving the marginal target distribution. Random generation sampled every raw coordinate independently from a uniform distribution spanning its observed corpus range. Vector generation drew raw coordinates with replacement across systems and added Gaussian noise scaled to each coordinate’s population standard deviation. Falsified datasets were generated once under base seed 42. Both frozen and retrained models tested degradation via one-sided Wilcoxon signed-rank tests ($H_0 : \theta \geq 0$ versus $H_1 : \theta < 0$) with Holm adjustment across the six comparison pairs.

Ablation assessed structural dependence by comparing the log-additive decomposition against four alternative input specifications. Two simple specifications fit a single-stage model at $\tau_C = 0.99$ using invariants only (x') or signatures only (z'). Two complex specifications tested alternative couplings: joint features combined x' and z' into a single-stage model, while interaction terms augmented the second stage with all 147 products of raw coordinates $x_j z_k$, standardized within each training fold. All ablations were refitted using the standard estimator protocols. One-sided signed-rank tests evaluated non-inferiority of the log-additive decomposition [68] ($H_0 : \theta \geq \delta$ versus $H_1 : \theta < \delta$) on EI and CI with Holm adjustments applied separately within simple and complex specification pairs.

Implementation and Reproducibility

All computational pipelines, graph algorithms, and statistical evaluations were implemented in Python 3.11 using deterministic, versioned environments [69, 70]. Graph representations and invariant calculations used python-igraph 1.0.0 [71, 72] with NumPy 2.3.5 [73, 74], SciPy 1.16.3 [75, 76], and pandas 3.0.0 [77, 78]. Urban street network construction used networkx 3.5 [79, 80] and OSMnx 2.0.6 (Supplementary Reference S22).

Benchmark temporal graph datasets were loaded using torch-geometric 2.7.0 [81, 82] and torch-geometric-temporal 0.56.2 [83, 84]. Observational records for gravitational wave and meteorological systems were retrieved via gwosc 0.8.1 and meteostat 1.7.6. Estimators used scikit-learn 1.8.0 [85, 86], sklearn-quantile 0.1.1 [87], XGBoost 3.1.1 [43, 88], and PyTorch 2.9.0 [89, 90]. Distance correlation was computed using dcor 0.7 [91], and scientific figures were generated with matplotlib 3.9.2 [92, 93]. Unspecified settings followed library defaults.

All stochastic procedures descended from base random seed 42 with repeat indices applied as fixed numerical offsets. Random permutations, cross-validation folds, and perturbation realizations originated from explicitly seeded generators. PyTorch deterministic algorithms operated in warn-only mode, with device execution defaulting to CUDA when available and CPU otherwise. The code repository, curated datasets, and environment specifications provide a reproducible workflow, subject to hardware-level floating-point variations.

Data Availability

All datasets were collated from publicly available sources. Access details are provided in Supplementary Table 1. The processed records used in all analyses are distributed with the code repository at: <https://github.com/nickkunz/network-capacity>.

Code Availability

All code for data processing, frontier estimation, evaluation, perturbation, falsification, and ablation, along with the notebooks that reproduce every figure and table are available at: <https://github.com/nickkunz/network-capacity>.

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Author Contributions

N. Kunz initiated and developed the concept, methods, implementation, testing, documentation, and manuscript. H. O. Gao initiated and supervised the development of the investigation and manuscript. The authors have agreed to the submitted version of this manuscript.

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Competing Interests

The authors declare no competing interests.

Additional Information

Supplementary information is available in Supplementary Tables 1–9 and Supplementary Note 1.

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Supplementary Tables

Supplementary Table 1. Corpus composition, substrate definitions, sources, and date provenance.

System	Domain	Discipline	Nodes	Edges	Event	Target	Time (Days)	Source	Notes
<i>gwosc</i>	Earth & Physical Sciences	Cosmology	Gravitational-wave detectors (H1, L1, V1, K1, G1).	Complete graph among detectors.	GWTC event records from any GWTC catalog attributed to at least one corpus detector, including the marginal and auxiliary catalogs.	Maximum daily records $r_{\max} = 3$, $y^* = 1.386$.	2015-09-14 to 2024-01-09 ($n = 3,040$).	GWOSC event API (S1 , S24 , S25 , S26 , S27 , S30).	Catalog articles published in 2019, 2021, 2023, and 2024. GWTC-4.0 first released 26 August 2025 and its results article was published in 2026. Fixed event cutoff 2024-01-09.
<i>river</i>	Earth & Physical Sciences	Hydrology	USGS stream gauges on the Colorado River (HUC 15010001, 15010002, 15010005).	Non-crossing path among 6-nearest gauges.	Peaks of discharge (parameter 00060) above the site-wise 99th percentile.	Maximum daily discharge-peak exceedances $r_{\max} = 9$, $y^* = 2.303$.	2014-11-12 to 2023-08-01 ($n = 3,185$).	USGS NWIS site and instantaneous values services (S2).	NWIS dataset record published 1994. API query cutoff 2024-12-31.
<i>auger</i>	Earth & Physical Sciences	Physics	Pierre Auger Observatory surface-detector stations.	Reciprocal 6-nearest surface-detector neighbours within $1.1 \times$ array spacing (1,500 m and 750 m arrays).	Surface-detector air-shower events.	Maximum daily air-shower events $r_{\max} = 66$, $y^* = 4.205$.	2004-01-03 to 2018-09-01 ($n = 5,356$).	Auger Open Data station map and Zenodo record 10488964 (S3).	Open Data version 3 released 2024. Observations span 2004-2018.
<i>seismic</i>	Earth & Physical Sciences	Geology	IRIS IU network stations.	Complete graph among stations.	Reviewed USGS events, 2023.	Maximum daily reviewed events $r_{\max} = 1,129$, $y^* = 7.030$.	2023-01-01 to 2023-12-30 ($n = 364$).	IRIS FDSN station service (S4) and USGS FDSN event service (S5).	GSN dataset record published 1988. ComCat record published 2017. Event-query cutoff 2023-12-31.
<i>rain</i>	Earth & Physical Sciences	Meteorology	Meteostat hourly stations in Laos.	Delaunay triangulation of station coordinates.	Hours with precipitation > 1 mm.	Maximum daily wet hours $r_{\max} = 46$, $y^* = 3.850$.	2024-01-12 to 2024-03-30 ($n = 79$).	Meteostat (S6).	Mutable service accessed 1 September 2026. Query cutoff 2024-03-31.
<i>celegans</i>	Life Sciences & Medicine	Biology	<i>C. elegans</i> developmental stages.	Known stage-to-stage transitions (normal, dauer, and recovery pathways), entered undirected.	Developmental transitions at published mean times.	Maximum daily developmental transitions $r_{\max} = 8$, $y^* = 2.197$.	Days 0-33 ($n = 34$).	Cassada & Russell, <i>Dev. Biol.</i> 46 , 326-342 (1975) (S7), and Golden & Riddle, <i>Dev. Biol.</i> 102 , 368-378 (1984) (S8).	Timing studies published 1975 and 1984. Days are developmental-age indices, not calendar dates.

Supplementary Table 1. Corpus composition and provenance (*continued*).

System	Domain	Discipline	Nodes	Edges	Event	Target	Time (Days)	Source	Notes
<i>crop</i>	Life Sciences & Medicine	Ecology	Crops and pollinator taxa.	Observed crop-pollinator interactions.	Daily flower-visitor census visits from primary study records.	Maximum daily census visits $r_{\max} = 1,266$, $y^* = 7,144$.	2004-05-07 to 2012-05-29 ($n = 2,945$). (<i>S9</i>).	CropPol repository (Rader et al. 2004–2012).	Database article published 2022. Observations span 2004–2012.
<i>chickenpox</i>	Life Sciences & Medicine	Epidemiology	Hungarian counties.	County adjacency from the Chickenpox dataset.	Weekly county case counts summed across counties, divided by 7 to a daily average.	Maximum daily-average cases $r_{\max} = 378$, $y^* = 5,938$.	2005-01-03 to 2014-12-29 ($n = 3,648$).	Rozemberczki et al. 2021. Chickenpox dataset and UCI Hungarian chickenpox cases (<i>S10</i>).	UCI dataset released 2021. Observations span 2005–2014. Source reports weekly totals, converted to daily averages.
<i>faers</i>	Life Sciences & Medicine	Pharmacology	Medicinal products and MedDRA reaction terms co-reported with imatinib.	Drug-reaction co-occurrence.	Unique FAERS case receipts.	Maximum daily case receipts $r_{\max} = 45$, $y^* = 3,829$.	2004-01-23 to 2025-12-31 ($n = 8,014$).	openFDA drug-event API, query IMATINIB (<i>S11</i>).	Mutable API accessed 1 September 2026. Case-receipt cutoff 2025-12-31.
<i>epilepsy</i>	Life Sciences & Medicine	Neurology	19 scalp electrodes (10–20 system).	Anatomical neighbours on the 10–20 montage.	Annotated seizures in CHB-MIT (chb01–chb24).	Maximum daily annotated seizures $r_{\max} = 27$, $y^* = 3,332$.	Relative recording days 0–76 ($n = 77$).	PhysioNet CHB-MIT (<i>S12</i>).	PhysioNet version 1.0.0 released 2010. Calendar dates are unavailable, so recording order is expressed as relative days.
<i>federal</i>	Trade & Institutions	Government	Awarding agencies and recipients.	Agency–recipient contract pairs.	Waterfowl-related contract start dates.	Maximum daily contract starts $r_{\max} = 17$, $y^* = 2,890$.	2011-01-19 to 2024-12-30 ($n = 5,095$).	USAspending API, keyword waterfowl (<i>S13</i>).	Mutable API accessed 1 September 2026. Award-start cutoff 2024-12-31.
<i>mooc</i>	Trade & Institutions	Education	Students and course actions.	Complete bipartite graph between students and actions.	Student–action timestamps.	Maximum daily student actions $r_{\max} = 18,213$, $y^* = 9,810$.	Days 0–29 ($n = 30$).	SNAP act-mooc (<i>S14</i>).	Dataset article published 2019. Timestamps reindexed to relative days.
<i>amazon</i>	Trade & Institutions	Commerce	Reviewers and book products.	Reviewer–product review pairs.	Review timestamps.	Maximum daily reviews $r_{\max} = 37,293$, $y^* = 10,527$.	1996-05-20 to 2014-07-23 ($n = 6,639$).	SNAP Amazon product graph, Books ratings (<i>S15</i>).	Dataset article published 2015. Reviews span 1996–2014.
<i>bitcoin</i>	Trade & Institutions	Cryptocurrency	OTC users and integer rating values.	User-rating–user incidences over the complete Bitcoin OTC history.	Rating timestamps.	Maximum daily ratings $r_{\max} = 319$, $y^* = 5,768$.	2010-11-08 to 2016-01-25 ($n = 1,905$).	PyTorch Geometric BitcoinOTC (<i>S16</i>).	Dataset article published 2016. Ratings span 2010–2016.

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Supplementary Table 1. Corpus composition and provenance (*continued*).

System	Domain	Discipline	Nodes	Edges	Event	Target	Time (Days)	Source	Notes
<i>world</i>	Trade & Institutions	International Development	World Bank lending instruments and countries.	Instrument-country eligibility.	Project board-approval dates.	Maximum daily project approvals $r_{\max} = 11$, $y^* = 2.485$.	2015-12-21 to 2024-12-20 ($n = 3,288$).	World Bank projects and country APIs (S17).	Mutable APIs accessed 1 September 2026. Approval-year cutoff 2024.
<i>wiki</i>	Technology & Information	Knowledge Graphs	Wikipedia mathematics pages.	Page-page relations from WikiMaths snapshots.	Daily summed page activity.	Maximum daily summed page activity $r_{\max} = 1,323,743$, $y^* = 14.096$.	2019-03-16 to 2021-03-15 ($n = 731$).	UCI Wikipedia Math Essentials and PyTorch Geometric Temporal WikiMaths (S18).	UCI dataset released 2021. Activity spans 2019–2021.
<i>jodie</i>	Technology & Information	Interaction Networks	Wikipedia users and pages.	User-page interactions.	Interaction timestamps.	Maximum daily interactions $r_{\max} = 6,087$, $y^* = 8.714$.	Days 0–30 ($n = 31$).	PyTorch Geometric JODIE, Wikipedia (S14).	Dataset article published 2019. Timestamps reindexed to relative days.
<i>overflow</i>	Technology & Information	Online Forums	Stack Overflow users and calendar days.	Complete bipartite graph between users and days.	Comment, answer, and other timed interactions.	Maximum daily timed interactions $r_{\max} = 52,446$, $y^* = 10.868$.	2008-08-01 to 2016-03-06 ($n = 2,775$).	SNAP sx-stackoverflow (S19).	Dataset article published 2017. Interactions span 2008–2016.
<i>email</i>	Technology & Information	Email Exchanges	European research-institution email accounts and days.	Complete bipartite graph between accounts and days.	Email timestamps.	Maximum daily emails $r_{\max} = 2,270$, $y^* = 7.728$.	Days 0–803 ($n = 804$).	SNAP email-Eu-core-temporal (S19).	Dataset article published 2017. Timestamps reindexed to relative days.
<i>college</i>	Technology & Information	Social Networks	UC Irvine college-message users and days.	Complete bipartite graph between users and days.	Private-message timestamps.	Maximum daily private messages $r_{\max} = 2,678$, $y^* = 7.893$.	2004-04-15 to 2004-10-26 ($n = 195$).	SNAP CollegeMsg (S20).	Dataset article published 2009. Messages were recorded in 2004.
<i>idling</i>	Transportation & Infrastructure	Mobility	Halifax drive-network vertices from the unsimplified OSM graph.	Street segments (OSM drive graph, the unsimplified undirected).	Vehicle-idling records.	Maximum daily idling records $r_{\max} = 732$, $y^* = 6.597$.	2025-06-02 to 2025-09-02 ($n = 93$).	OpenStreetMap via OSMnx, query Halifax, Canada (S21 , S22), and local idling CSVs (S29).	Data preprint first posted 2024 and revised 2025. Observations are from 2025. OSMnx article published 2025.
<i>windmill</i>	Transportation & Infrastructure	Energy	Wind turbines.	Complete graph among turbines.	Raw-production on/off turbine transitions (threshold 10^{-6}).	Maximum daily turbine transitions $r_{\max} = 7$, $y^* = 2.079$.	Days 0–727 ($n = 728$).	PyTorch Geometric Temporal WindmillOutput Large [83] .	Loader article published 2021. Source supplies snapshot order rather than calendar dates.

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Supplementary Table 1. Corpus composition and provenance (*continued*).

System	Domain	Discipline	Nodes	Edges	Event	Target	Time (Days)	Source	Notes
<i>metrila</i>	Transportation & Infrastructure	Traffic	METR-LA loop detectors.	Detector adjacency from METR-LA snapshots.	Onset of congestion (raw speed at or below the detector-wise 1st percentile).	Maximum daily congestion onsets $r_{\max} = 675$, $y^* = 6.516$.	2012-03-01 to 2012-06-27 ($n = 119$).	PyTorch Geometric Temporal METRLA [83] (S23).	Benchmark article published 2018 and loader article 2021. Sensor observations are from 2012.
<i>pemsbay</i>	Transportation & Infrastructure	Traffic II	PEMS-BAY loop detectors.	Detector adjacency from PEMS-BAY snapshots.	Onset of congestion (raw speed at or below the detector-wise 1st percentile).	Maximum daily congestion onsets $r_{\max} = 634$, $y^* = 6.454$.	2017-01-01 to 2017-06-29 ($n = 180$).	PyTorch Geometric Temporal PEMS-BAY [83] (S23).	Benchmark article published 2018 and loader article 2021. Sensor observations are from 2017.
<i>monteideo</i>	Transportation & Infrastructure	Ridership	Monteideo bus stops.	Stop adjacency from MonteideoBus snapshots.	Hours with inflow above the global 99th percentile.	Maximum daily exceedance hours $r_{\max} = 256$, $y^* = 5.549$.	Days 0–29 ($n = 30$).	PyTorch Geometric Temporal MonteideoBus [83] (S28).	Data recorded in October 2020. Loader article published 2021. Dates represented as relative days.

Each row is one N3S. Nodes and edges describe the undirected substrate used for graph invariants. Events describe the observation stream aggregated to the series r_t of Eq. (2). Metrila, pemsbay, monteideo, windmill, and wiki supply regular snapshots that aggregate to complete daily series. Chickenpox supplies weekly county totals, each converted to its daily average and entered as 1 series value at its report date. For the remaining 19 systems, the series contains only days with at least 1 recorded event. Observation windows are the native recording periods, given as start–end dates or relative day indices, with the calendar span in days (n) from the first to the last recorded interval. Windows were not standardized in duration, and each empirical maximum is conditional on its native window. Targets give the realized maximum $r_{\max}$ and $y^* = \ln(1 + r_{\max})$ from Eq. (3). Chickenpox uses the daily average entered at the report date, and the remaining 24 systems use the daily event count. Observation dates identify the records analyzed and differ from source release or publication dates. Source notes preserve versions, relative-date conventions, and fixed analysis cutoffs. A publication year outside the observation window is therefore expected. Source citations S1–S30 refer to the [Supplementary References](#) below.

Supplementary Table 2. Domain transfer and baseline frontier performance.

Evaluation	Method	Median EI [Q1–Q3]	VR	MV	MS
Transfer	Domain (LOGO-CV)	0.76 [0.73–0.77]	0.04	0.21	6.93
	Discipline (LOGO-CV)	0.73 [0.71–0.75]	0.01	0.01	5.69
Baseline	Random (5-Fold-CV)	0.77 [0.75–0.77]	0.03	0.12	6.54
	Random (10-Fold-CV)	0.77 [0.75–0.77]	0.02	0.03	5.68

Each regime was evaluated using 9 estimators and 30 seeds (42–71). Entries are medians across estimator-level means over folds or groups and repeats. Brackets give the first and third quartiles across estimators. Groups, folds, repeats, and estimators received equal weight. LOGO-CV partitions were fixed across repeats, while random partitions were reshuffled. EI is the Efficiency Index, VR the violation rate, MV the mean exceedance among violators within a fold (0 when none occurred), and MS the mean non-negative slack across evaluated systems. MV and MS are in natural-log target units. These are descriptive summaries.

Supplementary Table 3. Equivalence of resampling regimes to domain-level LOGO-CV.

Domain Transfer Equivalence						
Evaluation	Median ΔEI	Rank-biserial r	TOST p	Holm-adj. p	Sig.	Eq.
Discipline (LOGO-CV)	0.00	0.04	0.008	0.023	*	Yes
Random (5-Fold-CV)	0.01	0.14	0.042	0.042	*	Yes
Random (10-Fold-CV)	0.00	0.20	0.012	0.024	*	Yes
Prediction Consensus Equivalence						
Evaluation	Median ΔCI	Rank-biserial r	TOST p	Holm-adj. p	Sig.	Eq.
Discipline (LOGO-CV)	0.00	-0.08	0.026	0.026	*	Yes
Random (5-Fold-CV)	0.01	0.04	0.005	0.010	*	Yes
Random (10-Fold-CV)	0.01	0.03	0.001	0.004	**	Yes

Averaged out-of-fold predictions from each regime were rescored within the same 5 broad domains. Each comparison uses 45 paired estimator-by-domain comparisons from 9 estimators and 5 domains, summarized over 30 seeds, with shifts defined as comparison regime minus Domain LOGO-CV. EI is the Efficiency Index, and CI is the Consensus Index between predictions and observed maxima. The rank-biserial effect r uses unshifted paired differences and has the same direction. p is the unadjusted TOST value, taken as the larger of the 2 one-sided Wilcoxon p values. p_H is Holm-adjusted across the 3 regime comparisons separately for each outcome. Sig. gives the Holm-adjusted significance code: *** for $p_H < 0.001$, ** for $p_H < 0.01$, and * for $p_H < 0.05$. Eq. indicates rejection of both TOST nulls after adjustment at 0.05, within IQR margins $\delta_{EI} = 0.01$ and $\delta_{CI} = 0.03$. These margins were calculated from model-level Domain LOGO-CV means after each model was averaged equally across domains. Shifts rounding to zero are displayed as 0.00. The 45 comparisons share estimators, training systems, and domains, so inference is conditional on this panel.

Supplementary Table 4. Maximum-intensity perturbation equivalence under a frozen frontier.

Domain Transfer							
Perturbation	Method	Median ΔEI	Rank-biserial r	TOST p	Holm-adj. p	Sig.	Eq.
Network	Sample	0.00	0.06	< 0.001	< 0.001	***	Yes
	Rewire	0.00	-0.44	< 0.001	< 0.001	***	Yes
	Densify	0.00	-0.49	< 0.001	< 0.001	***	Yes
Invariants	Jitter	0.00	-0.15	< 0.001	< 0.001	***	Yes
	Noise	0.00	-0.57	< 0.001	< 0.001	***	Yes
	Subset	0.00	-0.15	< 0.001	< 0.001	***	Yes
Process	Bootstrapping	0.01	0.26	< 0.001	< 0.001	***	Yes
	Scaling	0.00	-0.15	< 0.001	< 0.001	***	Yes
	Smoothing	-0.02	-0.62	< 0.001	< 0.001	***	Yes
Signatures	Jitter	0.00	-0.41	< 0.001	< 0.001	***	Yes
	Noise	0.00	-0.61	< 0.001	< 0.001	***	Yes
	Subset	-0.01	-0.49	< 0.001	< 0.001	***	Yes
Prediction Consensus							
Perturbation	Method	Median ΔCI	Rank-biserial r	TOST p	Holm-adj. p	Sig.	Eq.
Network	Sample	0.00	-0.56	< 0.001	< 0.001	***	Yes
	Rewire	0.00	-0.35	< 0.001	< 0.001	***	Yes
	Densify	0.00	-0.50	< 0.001	< 0.001	***	Yes
Invariants	Jitter	0.00	-0.53	< 0.001	< 0.001	***	Yes
	Noise	0.00	-0.43	< 0.001	< 0.001	***	Yes
	Subset	0.00	-0.58	< 0.001	< 0.001	***	Yes
Process	Bootstrapping	-0.04	-0.83	< 0.001	< 0.001	***	Yes
	Scaling	-0.02	-0.57	< 0.001	< 0.001	***	Yes
	Smoothing	-0.06	-0.64	< 0.001	< 0.001	***	Yes
Signatures	Jitter	-0.01	-0.47	< 0.001	< 0.001	***	Yes
	Noise	-0.03	-0.75	< 0.001	< 0.001	***	Yes
	Subset	-0.06	-0.87	< 0.001	< 0.001	***	Yes

EI is the Efficiency Index and CI is the Consensus Index between predictions and observed maxima. Shifts are medians of paired differences (perturbed minus original), with positive values indicating higher scores under perturbation. The rank-biserial effect r uses unshifted paired differences and has the same direction. Shifts and rank-biserial effects are rounded to two decimals, with shifts rounding to zero displayed as 0.00. The frozen domain-level LOGO-CV protocol uses 45 paired estimator-by-domain comparisons (9 estimators and 5 domains), summarized over 30 seeds. Only the maximum tested intensity for each method enters the test. Sample denotes node removal, and subset denotes coordinate masking. p is the unadjusted TOST value, taken as the larger of the two one-sided Wilcoxon p values. p_H is Holm-adjusted across the 12 methods separately for each outcome. Both p-value columns are reported to three decimals, with values below 0.001 shown as < 0.001. Sig. gives the Holm-adjusted significance code: *** for $p_H < 0.001$, ** for $p_H < 0.01$, and * for $p_H < 0.05$. Eq. indicates rejection of both TOST nulls after adjustment at 0.05, within IQR margins $\delta_{EI} = 0.09$ and $\delta_{CI} = 0.15$. These margins are derived from original-condition variability. The 45 comparisons share estimators, training systems, and domains, so inference is conditional on this panel.

Supplementary Table 5. Falsification effects under frozen and retrained frontiers.

Domain Transfer							
Falsification	Method	Median ΔEI	Rank-biserial r	One-sided p	Holm-adj. p	Sig.	Diff.
Frozen	Target Permutation	-0.02	-0.43	0.005	0.005	**	Yes
	Random Generation	-0.05	-0.89	< 0.001	< 0.001	***	Yes
	Vector Generation	-0.03	-0.70	< 0.001	< 0.001	***	Yes
Retrain	Target Permutation	-0.08	-0.49	0.002	0.003	**	Yes
	Random Generation	-0.03	-0.52	0.001	0.003	**	Yes
	Vector Generation	-0.03	-0.60	< 0.001	< 0.001	***	Yes
Prediction Consensus							
Falsification	Method	Median ΔCI	Rank-biserial r	One-sided p	Holm-adj. p	Sig.	Diff.
Frozen	Target Permutation	-0.24	-0.92	< 0.001	< 0.001	***	Yes
	Random Generation	-0.24	-0.92	< 0.001	< 0.001	***	Yes
	Vector Generation	-0.21	-0.90	< 0.001	< 0.001	***	Yes
Retrain	Target Permutation	-0.33	-0.98	< 0.001	< 0.001	***	Yes
	Random Generation	-0.21	-0.94	< 0.001	< 0.001	***	Yes
	Vector Generation	-0.28	-0.98	< 0.001	< 0.001	***	Yes

EI is the Efficiency Index and CI is the Consensus Index between predictions and observed maxima. Shifts are medians of paired differences (falsified minus original), with negative values indicating lower scores under falsification. The rank-biserial effect r is computed from the unshifted paired differences and has the same direction. Each comparison uses 45 paired estimator-by-domain comparisons under domain-level LOGO-CV, summarized over 30 seeds. Frozen frontiers retain original-data fits, while retrained frontiers refit the pipeline on falsified data. p is the unadjusted one-sided Wilcoxon signed-rank value. p_H is Holm-adjusted across the six method-by-frontier comparisons separately for each outcome. Both p-value columns are reported to three decimals, with values below 0.001 shown as < 0.001. Sig. gives the Holm-adjusted significance code: *** for $p_H < 0.001$, ** for $p_H < 0.01$, and * for $p_H < 0.05$. Diff. indicates $p_H < 0.05$ and a negative median shift, supporting inferiority to the original condition. The falsified corpora are corpus-level negative controls. Contrasts share estimators, training systems, and domains, so inference is conditional on this panel.

Supplementary Table 6. Non-inferiority of the log-additive decomposition to alternative specifications.

Domain Transfer						
Ablation	Method	Median ΔEI	Rank-biserial r	One-sided p	Holm-adj. p	Sig. NI.
Simple	Invariants Only	-0.01	-0.46	< 0.001	< 0.001	*** Yes
	Signatures Only	0.00	-0.09	< 0.001	< 0.001	*** Yes
Complex	Joint Features	-0.01	-0.46	< 0.001	< 0.001	*** Yes
	Interaction Terms	-0.01	-0.38	< 0.001	< 0.001	*** Yes
Prediction Consensus						
Ablation	Method	Median ΔCI	Rank-biserial r	One-sided p	Holm-adj. p	Sig. NI.
Simple	Invariants Only	-0.30	-0.87	< 0.001	< 0.001	*** Yes
	Signatures Only	-0.09	-0.54	< 0.001	< 0.001	*** Yes
Complex	Joint Features	-0.26	-0.69	< 0.001	< 0.001	*** Yes
	Interaction Terms	-0.08	-0.71	< 0.001	< 0.001	*** Yes

EI is the Efficiency Index and CI is the Consensus Index between predictions and observed maxima. Shifts are medians of paired differences (alternative specification minus log-additive decomposition). Negative values favor the log-additive decomposition. The rank-biserial effect r uses unshifted differences, so positive values indicate higher scores under the alternative independently of the tested margin. Each comparison uses 45 paired estimator-by-domain comparisons under retrained domain-level LOGO-CV, summarized over 30 seeds. The IQR non-inferiority margins are $\delta_{EI} = 0.07$ and $\delta_{CI} = 0.14$, calculated from variability under the log-additive decomposition. The one-sided tests assess $H_0 : \Delta \geq \delta$ against $H_1 : \Delta < \delta$. p is the unadjusted one-sided Wilcoxon signed-rank value. p_H is Holm-adjusted separately within the two simple and the two complex specifications for each outcome. Both p-value columns are reported to three decimals, with values below 0.001 shown as < 0.001. Sig. gives the Holm-adjusted significance code: *** for $p_H < 0.001$, ** for $p_H < 0.01$, and * for $p_H < 0.05$. NI indicates support for non-inferiority of the log-additive decomposition when $p_H < 0.05$ and the median shift is below the margin. This decision concerns advantages of at least the stated margin. It does not establish inferiority of simple specifications. Shifts rounding to zero are displayed as 0.00. Contrasts share estimators, training systems, and domains, so inference is conditional on this panel.

Supplementary Table 7. 21 Classical Graph Invariants by Category – Definitions, Formulae, and Asymptotic Time Complexity.

Category	Invariant	Definition	Formula	Complexity
Simple	1. Number of Nodes	Number of vertices $ V $	$n = V $	$O(1)$
	2. Number of Edges	Number of edges $ E $	$m = E $	$O(1)$
	3. Articulation Points	Vertices whose removal splits G	$ \{v : \omega(G - v) > \omega(G)\} $	$O(n + m)$
	4. Bridges	Edges whose removal splits G	$ \{e : \omega(G - e) > \omega(G)\} $	$O(n + m)$
Cohesion	5. Diameter	Longest shortest-path length $\dagger$	$\max_v \epsilon(v)$	$O(nm)$
	6. Radius	Smallest vertex eccentricity $\dagger$	$\min_v \epsilon(v)$	$O(nm)$
	7. Degeneracy	Largest k with a k -core	$k^* = \max_v c(v)$	$O(n + m)$
	8. k -Core Size	Vertices in the core at k	$ \{v : c(v) = k^*\} $	$O(n + m)$
Extremal	9. Maximum Degree	Highest vertex degree	$\max_v d_v$	$O(n)$
Statistical	10. Degree Variance	Spread of degrees	$n^{-1} \sum_v (d_v - \bar{d})^2$	$O(n)$
	11. Degree Entropy	Entropy of degree distribution	$-\sum_k p_k \ln p_k$	$O(n)$
	12. Joint Degree Entropy	Entropy of edge degree pairs	$-\sum_{k \leq l} q_{kl} \ln q_{kl}$	$O(m)$
	13. Degree Skewness	Asymmetry of degree distribution	$\frac{\sqrt{n(n-1)}}{n-2} g_1$	$O(n)$
	14. Degree Kurtosis	Tail weight of degree distribution	$\frac{(n-1)[(n+1)g_2+6]}{(n-2)(n-3)}$	$O(n)$
	15. Global Clustering Coefficient	Fraction of closed triples	$3N_{\triangle}/N_{\wedge}$	$O(nd_{\max}^2)$
Spectral	16. Assortativity	Correlation of endpoint degrees	$\text{corr}_{\{u,v\} \in E}(d_u, d_v)$	$O(m)$
	17. Normalized Laplacian 2nd Moment	Diffusion intensity	$\text{tr}(L^2)/n$	$O(m)$
	18. Normalized Laplacian 3rd Moment	Diffusion asymmetry	$\text{tr}(L^3)/n$	$O(m^2)$
	19. Random Walk 3rd Moment	Three-step recurrence	$\text{tr}(P^3)/n$	$O(m^2)$
	20. Random Walk 4th Moment	Four-step recurrence	$\text{tr}(P^4)/n$	$O(m^2)$
	21. Adjacency 4th Moment	Density of four-length walks	$\text{tr}(A^4)/n$	$O(m^2)$

Admission criteria were prespecified. Each invariant was required to be (i) formally invariant, defined purely by the graph and preserved under relabeling, with no dependence on embeddings, weights, external parameters, or arbitrary thresholds, (ii) exactly and deterministically computable at the largest corpus scales, and (iii) globally informative, a whole-graph descriptor with non-trivial variation across systems. $\dagger$ Diameter and radius were computed on the largest connected component. Spectral moments were computed as trace polynomials of sparse matrices, avoiding eigendecomposition at corpus scale, and all values were constrained finite. Here $n = |V|$ and $m = |E|$. In the formulae, d_v is the degree of v , $\bar{d}$ the mean degree, $\mu_j = n^{-1} \sum_v (d_v - \bar{d})^j$, $g_1 = \mu_3/\mu_2^{3/2}$, $g_2 = \mu_4/\mu_2^2 - 3$, ω the number of connected components, $\epsilon(v)$ the eccentricity of v in hops, $c(v)$ its core number, p_k the fraction of vertices of degree k , q_{kl} the fraction of edges with unordered endpoint degrees $\{k, l\}$, and $N_{\triangle}$ and $N_{\wedge}$ the numbers of triangles and connected triples. The matrix A is the adjacency matrix, $\hat{A} = D^{-1/2} A D^{-1/2}$ is taken over non-isolated vertices, $L = I - \hat{A}$, and $P = D^{-1} A$. Isolated vertices count toward n and add 0 to $\text{tr}(L^k)$, logarithms are natural, assortativity counts both orientations of each edge, and degeneracy and k -core size use the full graph. Undefined assortativity and clustering, and the skewness and kurtosis of equal degrees or of $n < 3$ and $n < 4$ vertices, respectively, were set to 0. The 4 complete bipartite substrates used closed-form values of these formulae. Feature names correspond to the released code. Invariant references: [94–104].

Supplementary Table 8. 7 Process Signatures by Property – Definitions, Formulae, and Asymptotic Time Complexity.

Property	Signature	Definition	Formula	Complexity
Dependence	1. Lag-1 Autocorrelation	Short-term sequential dependence	$\frac{\sum_{i=1}^{t-1} \tilde{r}_i \tilde{r}_{i+1}}{\sum_{i=1}^t \tilde{r}_i^2}$	$O(t)$
	2. Hurst Exponent	Long-range dependence	$\ln(R/s)/\ln t$	$O(t)$
Dispersion	3. Coefficient of Variation	Scale-free dispersion	$s/\bar{r}$	$O(t)$
	4. Fano Factor	Deviation from Poisson dispersion	$s^2/\bar{r}$	$O(t)$
Volatility	5. Normalized Successive Difference	Local temporal volatility	$\frac{1}{(t-1)\bar{r}} \sum_{i=1}^{t-1} r_{i+1} - r_i $	$O(t)$
Trend	6. Trend Coefficient	Systematic non-stationarity	$\text{corr}(r_i, i)$	$O(t)$
Asymmetry	7. Count Skewness	Distributional asymmetry	$\frac{1}{t} \sum_{i=1}^t (\tilde{r}_i/s)^3$	$O(t)$

Admission criteria were prespecified. Each signature was required to be (i) defined purely by the count series, with no dependence on external parameters or arbitrary thresholds, (ii) exactly and deterministically computable, and (iii) globally informative, a whole-series descriptor with non-trivial variation across systems. Signatures were computed from the daily count series r_t of Eq. (2), where t denotes the series length, and all values were constrained finite. In the formulae, $r_1, \dots, r_t$ is the series in chronological order, $\bar{r}$ is its mean, $\tilde{r}_i = r_i - \bar{r}$, and s is its sample standard deviation with $t - 1$ in the denominator. The Hurst exponent is the single-window rescaled-range estimate over the full series, clipped to $[0, 1]$, with $R = \max_i Y_i - \min_i Y_i$ the range of the cumulative sums $Y_i = \sum_{j \leq i} \tilde{r}_j$. The trend coefficient is the Pearson correlation of each value with its position i . CV, Fano factor, and normalized successive difference are 0 when $\bar{r} \leq 0$. Lag-1 autocorrelation and trend are 0 when their denominators are at most 10^{-12} . Skewness is 0 when $t < 3$ or $s \leq 10^{-12}$, and the Hurst exponent is 0.5 when $t < 8$ or when R or s is at most 10^{-12} . Feature names correspond to the released code. Signature references: [105–113].

Supplementary Table 9. Stress-test protocol: interventions, settings, tracks, and inference.

Category	Method	Setting	Track
Perturbation (TOST equivalence at maximum intensity, $\delta_{EI} = 0.09$, $\delta_{CI} = 0.15$)			
Network	Rewire	Degree-preserving edge swaps, fraction 0.05–0.35 (max 0.35)	Frozen
	Densify	Uniform edge addition, fraction 0.05–0.35 (max 0.35)	Frozen
	Sample	Uniform node removal, fraction 0.05–0.35 (max 0.35)	Frozen
Invariants	Noise	Additive Gaussian noise, corpus-wide, 0.05–0.35 (max 0.35)	Frozen
	Jitter	Multiplicative jitter, 0.05–0.35 (max 0.35)	Frozen
	Subset	Random coordinate masking, retention 0.95–0.65 (max 0.65)	Frozen
Process	Bootstrapping	Bootstrap to destroy temporal order 0.05–0.35 (max 0.05)	Frozen
	Scaling	Power-law scaling, exponent 0.25–1.75 (max 0.25)	Frozen
	Smoothing	Rolling-mean smoothing, window 3–15 days (max 15)	Frozen
Signatures	Noise	Additive Gaussian noise, corpus-wide, 0.05–0.35 (max 0.35)	Frozen
	Jitter	Multiplicative jitter, 0.05–0.35 (max 0.35)	Frozen
	Subset	Random coordinate masking, retention 0.95–0.65 (max 0.65)	Frozen
Falsification (one-sided inferiority to the original condition, Holm-adjusted)			
Negative control	Target Permutation	Cross-domain derangement of y^* across systems with x and z unchanged	Frozen + Retrain
	Random Generation	Each feature redrawn uniformly within its observed corpus range	Frozen + Retrain
	Vector Generation	Feature bootstrap across systems plus Gaussian noise at pooled dispersion	Frozen + Retrain
Ablation (one-sided non-inferiority of the log-additive decomposition, $\delta_{EI} = 0.07$, $\delta_{CI} = 0.14$)			
Simple	Invariants Only	Frontier fitted on x' alone	Retrain
	Signatures Only	Frontier fitted on z' alone	Retrain
Complex	Joint Features	x' and z' concatenated into a single-stage fit	Retrain
	Interaction Terms	Log-additive decomposition augmented with all 147 pairwise products	Retrain

All settings used 30 model-fitting seeds, paired 1-to-1 with 30 independent realizations for stochastic interventions, and deterministic interventions used 30 model-fitting seeds over a single realization. Grids used 7 equally spaced values between the stated end points, and the scaling grid includes the identity $\gamma = 1$. Here s is the intensity, σ_j the sample standard deviation of raw coordinate j across the 25 systems, p the number of coordinates in x or z , ρ the retention, w the window, and $|T|$ the series length. For complete bipartite substrates, node removal used exact closed-form invariants, and rewiring and densification used the deterministic approximations of [Supplementary Note 1](#), with rewiring interpolated toward configuration-model estimates [114]. Falsified corpora were generated once under base seed 42 and served as corpus-level negative controls. The maximum tested intensity was the most severe setting, with severity equal to s or w for most methods, $1 - \rho$ for masking, $1 - s$ for bootstrapping, and $|\ln \gamma|$ for scaling. Inference used the 45 estimator-by-domain panel summarized over 30 seeds, with Holm adjustment within each test family.

Supplementary Note 1: Complete Bipartite Perturbations

The 4 complete bipartite substrates (college, email, mooc, overflow) were represented by their partition sizes m and n , with $N = m + n$ vertices and $M = mn$ edges, because their explicit graphs reach up to 7,200,000,000 edges. Node removal at intensity s deleted $\lfloor sN \rfloor$ vertices drawn uniformly without replacement, counted the removed vertices in each partition, and evaluated the closed forms of Supplementary Table 7 for the remaining complete bipartite graph. This computation is exact and used 30 realizations. Rewiring and densification used a deterministic degree-class approximation with 1 realization.

For degree classes c with degree d_c and N_c vertices, and an edge count M' , let

$$N = \sum_c N_c, \quad S_1 = \sum_c N_c d_c, \quad S_2 = \sum_c N_c d_c^2, \quad \bar{d} = \frac{S_1}{N}, \quad \kappa = \frac{S_2}{\max(S_1, 1)} - 1, \quad \bar{d}^{-1} = \frac{1}{N} \sum_c \frac{N_c}{\max(d_c, 1)}.$$

The approximation Φ assigned the following values and set articulation points, bridges, clustering, and assortativity to 0.

- Size: $|V| = N$, $|E| = M'$, and $d_{\max} = \max_c d_c$.
- Cohesion: diameter $D = \max\{2, \ln \max(N, 2) / \ln \kappa\}$ when $\kappa > 1$ and $D = N$ otherwise, radius $D/2$, degeneracy $k^* = \min(d_{\max}, \sqrt{2M'})$, and k -core size $\sum_c N_c \mathbf{1}[d_c \geq k^*]$.
- Degree statistics: variance $N^{-1} \sum_c N_c (d_c - \bar{d})^2$, bias-corrected skewness and excess kurtosis of the class-weighted degrees, degree entropy $-\sum_k p_k \ln p_k$ over degrees rounded to integers k with vertex fractions p_k , and joint degree entropy $-2 \sum_k q_k \ln q_k$ with stub fractions $q_k \propto k p_k$.
- Spectral moments: $\text{tr}(L^2)/N = 1 + 2M'/\max(S_1, 1)^2$, $\text{tr}(L^3)/N = 1 + (S_2 - S_1)^3/\max(NS_1^3, 1)$, $\text{tr}(P^3)/N = 6(S_2 - S_1)^2 \bar{d}^{-1}/[\max(6S_1, 1) \max(S_1, 1)]$, $\text{tr}(P^4)/N = N^{-1} \sum_c N_c [d_c S_2 / \max(S_1^2, 1) + \max(\kappa, 0) / \max(d_c, 1)]$, and $\text{tr}(A^4)/N = [S_2 + S_2^2 / \max(2S_1, 1)]/N$.

Rewiring at intensity s evaluated Φ on the classes of $K_{m,n}$, with degrees (n, m) , class sizes (m, n) , and $M' = M$, and interpolated each invariant as $(1 - \alpha)x_j + \alpha\Phi_j$ with $\alpha = 1 - (1 - s)^2$, where x_j is the exact $K_{m,n}$ value. Densification added $a = \min(\lfloor sM \rfloor, C)$ edges among the $C = \binom{m}{2} + \binom{n}{2}$ within-partition vertex pairs, so that each pair was added with probability $q = a/C$. It evaluated Φ with degrees $(n + q(m - 1), m + q(n - 1))$, class sizes (m, n) , and $M' = M + a$. It then multiplied the $K_{m,n}$ articulation points, bridges, and assortativity by $1 - q$ and set clustering to $\min\{c_0 + (1 - c_0)q, 1\}$, where c_0 is the $K_{m,n}$ clustering. All other substrates were perturbed as explicit graphs.

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